

# Week 10 Cheat Sheet

## 1. Problem Setup

We consider sequential decision making under uncertainty.

**State:**  $x_t$

**Action:**  $u_t$

**Cost:**  $\ell(x_t, u_t)$

**Dynamics:**  $P(x_{t+1}|x_t, u_t)$

**Goal:**

$$\min \mathbb{E} \left[ \sum_{t=0}^{\infty} \ell(x_t, u_t) \right]$$

## 2. Bellman Equation (Dynamic Programming)

$$J^*(x_t) = \min_{u_t} \mathbb{E} [\ell(x_t, u_t) + J^*(x_{t+1})]$$

**Interpretation:**

- Immediate cost + expected future cost
- Recursive decomposition of optimal control problem

## 3. Value Iteration (VI)

Iteratively update value function:

$$J_{k+1}(x) = \min_u \mathbb{E} [\ell(x, u) + J_k(x')]$$

**Repeat until convergence:**

$$J_k \rightarrow J^*$$

**Then extract policy:**

$$\pi^*(x) = \arg \min_u \mathbb{E} [\ell(x, u) + J^*(x')]$$

## 4. Policy Iteration (PI)

Alternates between evaluation and improvement.

### Step 1: Policy Evaluation

$$J^\pi(x) = \ell(x, \pi(x)) + \mathbb{E}[J^\pi(x')]$$

### Step 2: Policy Improvement

$$\pi_{\text{new}}(x) = \arg \min_u [\ell(x, u) + \mathbb{E}[J^\pi(x')]]$$

**Repeat:**

$$\pi \rightarrow \pi^*$$

## 5. Reinforcement Learning (RL)

**Key Difference:** Model is UNKNOWN

$$P(x'|x, u), \quad \ell(x, u) \text{ are unknown}$$

**Most problems:**

- Infinite horizon
- Learn through interaction

### RL Pipeline

1. **Learning:** collect data from environment
2. **Planning:** apply VI or PI
3. **Execution:** run learned policy

**Return:**

$$G = \sum_{t=0}^{\infty} \ell(x_t, u_t)$$

## 6. Temporal Difference (TD) Learning

**Key Idea:** Update value estimates using samples, without waiting for full trajectory.

### Monte Carlo (MC)

$$J(x) = \frac{1}{k} \sum_{i=1}^k \sum_{t=0}^T \ell(x_t, u_t)$$

**Problem:**

- Must wait until end of episode

### TD Learning

$$J(x_t) \leftarrow J(x_t) + \alpha [\ell(x_t, u_t) + J(x_{t+1}) - J(x_t)]$$

**TD Error:**

$$\delta_t = \ell(x_t, u_t) + J(x_{t+1}) - J(x_t)$$

**Advantages:**

- Online updates (no full trajectory needed)
- Faster convergence in practice

## 7. Key Comparisons

### DP vs RL

- DP: model known
- RL: model unknown

### VI vs PI

- VI: update value directly
- PI: alternate policy + value updates

## MC vs TD

- MC: wait until end
- TD: update step-by-step

## 8. Key Takeaways

- Bellman equation enables recursive optimization
- Policy Iteration = Evaluate + Improve
- RL learns from interaction (unknown model)
- TD learning enables online updates

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| Optimal control = DP (known model) + RL (unknown model) |
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